Interactions

Interactions

$$\begin{aligned}\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon\end{aligned}$$

We get a separate regression line for each species of penguin:

$$\text{bill length} = \begin{cases} \beta_0 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

> What are we assuming about the slopes of these different regression lines?

Interactions

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon$$

We get a separate regression line for each species of penguin:

$$\text{bill length} = \begin{cases} \beta_0 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Adelie} \\ \beta_0 + \beta_1 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Chinstrap} \\ \beta_0 + \beta_2 + \beta_3 \text{body mass} + \varepsilon & \text{species} = \text{Gentoo} \end{cases}$$

> What are we assuming about the slopes of these different regression lines?

That each line has the same slope

Interactions

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon$$

We get a separate regression line for each species of penguin:

$$\text{bill length} = \begin{cases} \beta_0 + \beta_3 \text{body mass} + \varepsilon & \text{if Adelie} \\ \beta_0 + \beta_1 + \beta_3 \text{body mass} + \varepsilon & \text{if Chinstrap} \\ \beta_0 + \beta_2 + \beta_3 \text{body mass} + \varepsilon & \text{if Gentoo} \end{cases}$$

> How can we modify our regression model to allow a different slope for each species?

Allowing the slope to be different

$$\text{bill length} = \begin{cases} \beta_0 + \beta_3 \text{body mass} + \varepsilon & \text{if Adelie} \\ \beta_0 + \beta_1 + (\beta_3 + \beta_4)\text{body mass} + \varepsilon & \text{if Chinstrap} \\ \beta_0 + \beta_2 + (\beta_3 + \beta_5)\text{body mass} + \varepsilon & \text{if Gentoo} \end{cases}$$

- Slope for Adelie penguins: β_3
- Slope for Chinstrap penguins: $\beta_3 + \beta_4$

difference in slopes for Chinstrap vs. Adelie penguins

- Slope for Gentoo penguins: $\beta_3 + \beta_5$

difference in slopes for Gentoo vs. Adelie

Allowing the slope to be different

$$\text{bill length} = \begin{cases} \beta_0 + \beta_3 \text{body mass} + \varepsilon & \text{if Adelie} \\ \beta_0 + \beta_1 + (\beta_3 + \beta_4)\text{body mass} + \varepsilon & \text{if Chinstrap} \\ \beta_0 + \beta_2 + (\beta_3 + \beta_5)\text{body mass} + \varepsilon & \text{if Gentoo} \end{cases}$$

Written more concisely:

$$\begin{aligned} \text{bill length} = {} & \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ & + \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ & + \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon \end{aligned}$$

Interactions

$$\begin{aligned}\text{bill length} = \beta_0 &+ \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ &+ \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ &+ \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon\end{aligned}$$

- The terms $\text{body mass} \cdot \text{IsChinstrap}$ and $\text{body mass} \cdot \text{IsGentoo}$ denote an *interaction* between body mass and species
- Interactions allow the slope for body mass to depend on species
- $\text{body mass} \cdot \text{IsChinstrap}$ literally means multiplication

Activity

Work on the class activity (handout).

Activity

$$
\begin{aligned}
\widehat{\text{bill length}} = 26.99 &+ 5.18\ \overset{0}{\text{IsChinstrap}} - 0.26\ \overset{1}{\text{IsGentoo}} \\
&+ 0.0032\ \text{body mass} \\
&+ 0.0013\ \text{body mass} \cdot \overset{0}{\text{IsChinstrap}} \\
&+ 0.0010\ \text{body mass} \cdot \overset{1}{\text{IsGentoo}}
\end{aligned}
$$

> What is the estimated bill length of a Gentoo penguin with a body mass of 5000g?

Activity

$$\begin{aligned}\widehat{\text{bill length}} = 26.99 &+ 5.18\ \text{IsChinstrap} - 0.26\ \text{IsGentoo} \\ &+ 0.0032\ \text{body mass} \\ &+ 0.0013\ \text{body mass} \cdot \text{IsChinstrap} \\ &+ 0.0010\ \text{body mass} \cdot \text{IsGentoo}\end{aligned}$$

> What is the estimated bill length of a Gentoo penguin with a body mass of 5000g?

$$26.99 - 0.26 + (0.0032 + 0.0010) \cdot 5000 = 47.73$$

Activity

$$
\begin{aligned}
\widehat{\text{bill length}} = 26.99 &+ 5.18\,\text{IsChinstrap} - 0.26\,\text{IsGentoo} \\
&+ 0.0032\,\text{body mass} \\
&+ 0.0013\,\text{body mass} \cdot \text{IsChinstrap} \\
&+ 0.0010\,\text{body mass} \cdot \text{IsGentoo}
\end{aligned}
$$

What is the estimated slope on body mass, for Chinstrap penguins?

0.0032 + 0.0013

Activity

$$\begin{aligned}\widehat{\text{bill length}} = 26.99 &+ 5.18\,\text{IsChinstrap} - 0.26\,\text{IsGentoo} \\ &+ 0.0032\,\text{body mass} \\ &+ 0.0013\,\text{body mass} \cdot \text{IsChinstrap} \\ &+ 0.0010\,\text{body mass} \cdot \text{IsGentoo}\end{aligned}$$

> What is the estimated slope on body mass, for Chinstrap penguins?

$0.0032 + 0.0013 = 0.0045$

Interactions

$$
\begin{aligned}
\text{bill length} = \beta_0 &+ \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\
&+ \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\
&+ \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon
\end{aligned}
$$

- The terms $\text{body mass} \cdot \text{IsChinstrap}$ and $\text{body mass} \cdot \text{IsGentoo}$ denote an *interaction* between body mass and species
- Interactions allow the slope for body mass to depend on species
- $\text{body mass} \cdot \text{IsChinstrap}$ literally means multiplication

> What are some advantages and disadvantages of adding interaction terms?

Interactions: pros and cons

Pros: Interactions can allow us to

- better capture the true relationships in the data
- better predict the response
- answer different research question (e.g., is there a difference in slope for different penguin species?)

Cons: Interactions make the model more complex

- often harder to interpret
- more variability in estimating the model

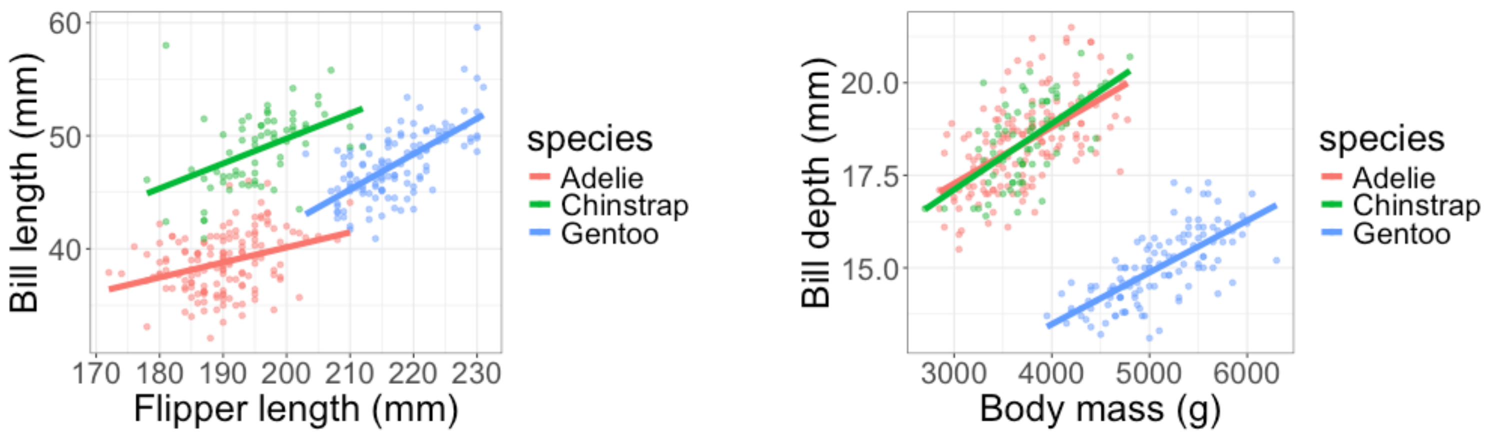
Do we need an interaction?

Ways to decide on interaction terms:

- Visualization
- Measuring predictive ability (e.g., R^2_{adj})
- Research question

Visualization

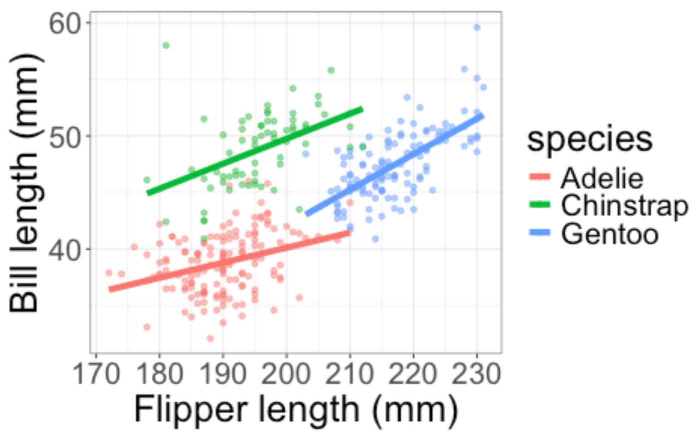
For interaction between categorical and quantitative predictors, we can plot the fitted regression lines for each group (with the interaction). Check whether the slopes look different.

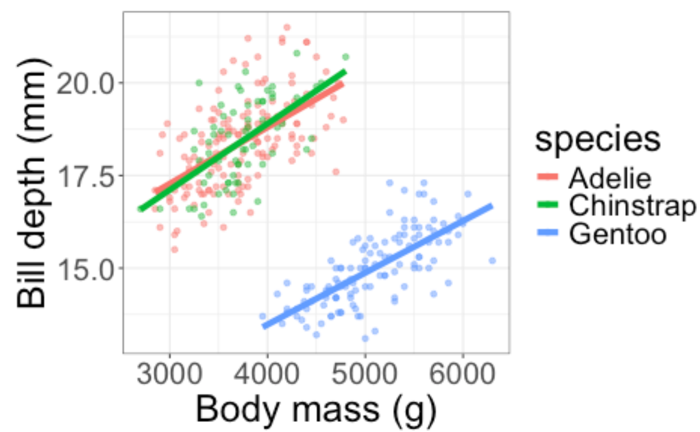


> Which plot suggests we might need an interaction?

Visualization

For interaction between categorical and quantitative predictors, we can plot the fitted regression lines for each group (with the interaction). Check whether the slopes look different.



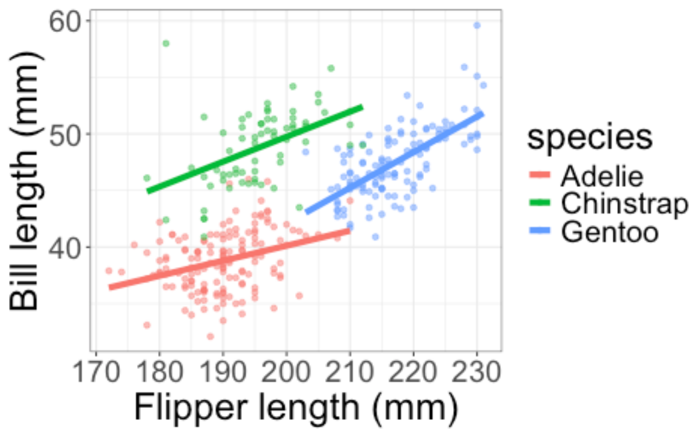


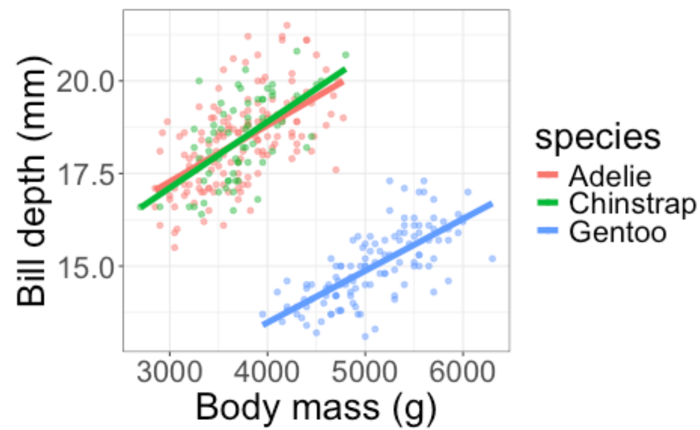
> Which plot suggests we might need an interaction?

The one on the left -- the slopes look different in the different groups

Measuring predictive ability

Compare R^2_{adj} with and without the interaction term.





With interaction (different slopes): $R^2_{adj} = 0.782$

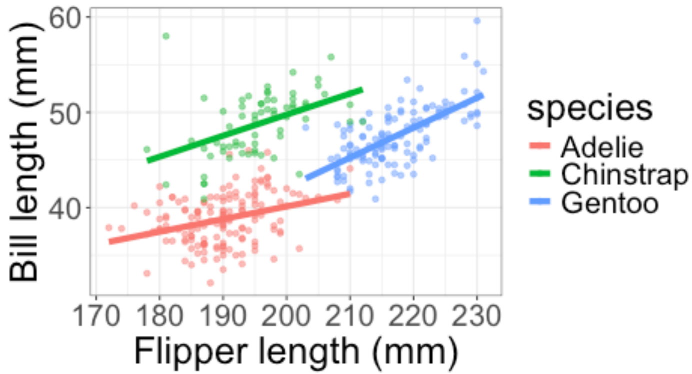
Without interaction (same slopes): $R^2_{adj} = 0.774$

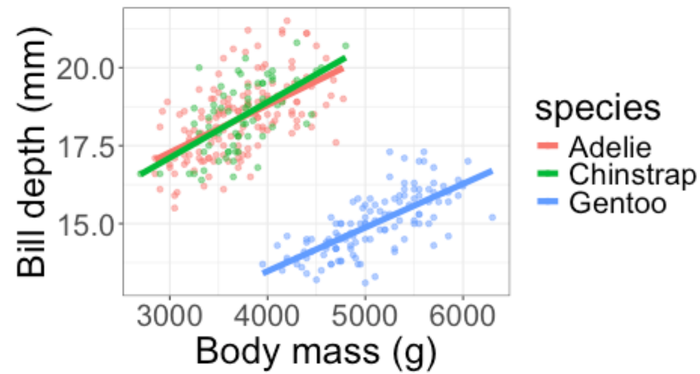
With interaction (different slopes): $R^2_{adj} = 0.8008$

Without interaction (same slopes): $R^2_{adj} = 0.8012$

R^2_{adj} decreases b/c SSE stays similar, but # parameters ↑

Measuring predictive ability





With interaction (different slopes): $R^2_{adj} = 0.782$

Without interaction (same slopes): $R^2_{adj} = 0.774$

With interaction (different slopes): $R^2_{adj} = 0.8008$

Without interaction (same slopes): $R^2_{adj} = 0.8012$

If the interaction only increases R^2_{adj} by a small amount, we might still prefer the simpler model.

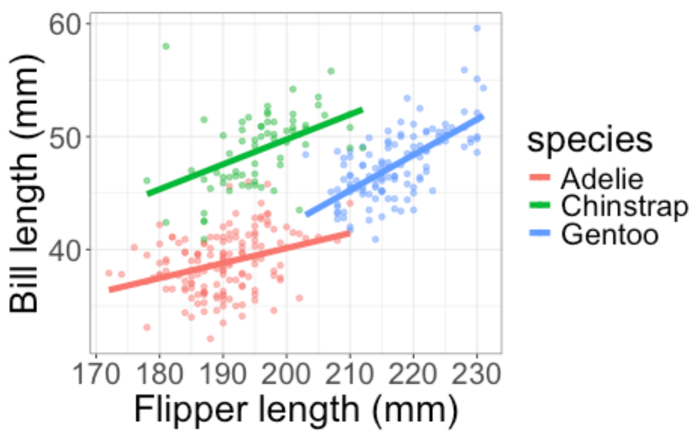
Considering research questions

Common types of research questions:

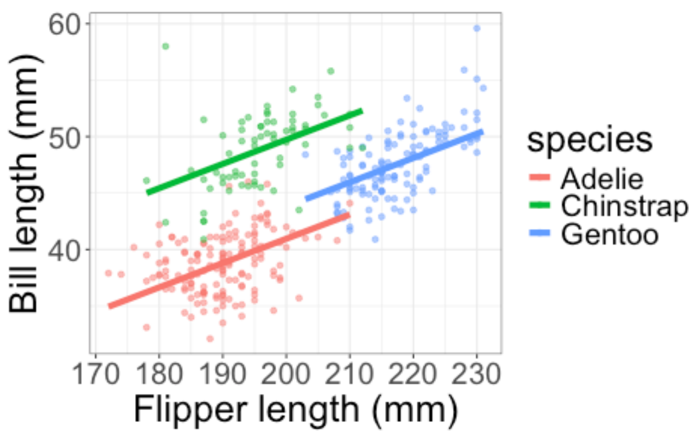
- Description (want to describe the overall relationship between 2 or more variables)
- Prediction (want to do a good job predicting the response)
- Inference (want to test hypotheses about the relationship)

Description

Example: What is the relationship between flipper length and bill length, accounting for species?



With interaction

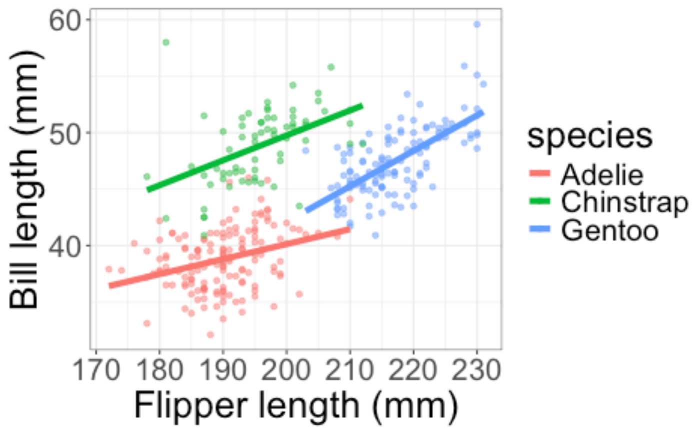


Without interaction

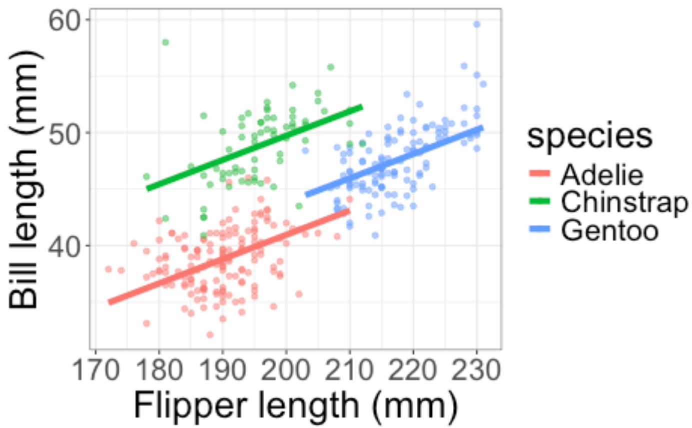
> Do we need the interaction to describe the general relationship between flipper length and bill length?

Inference

Example: Is there a difference in the slope on flipper length, for different species?



With interaction

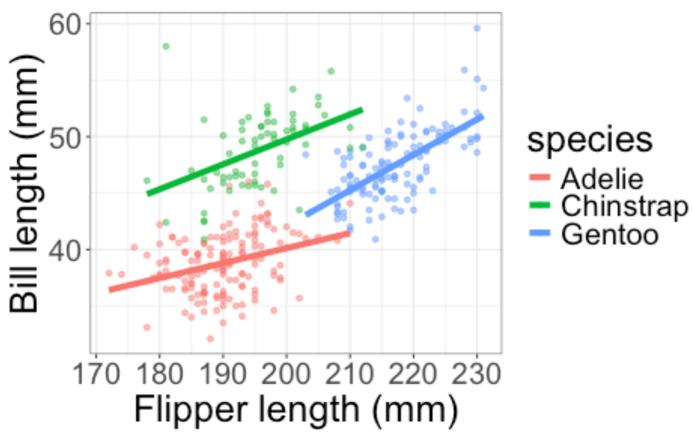


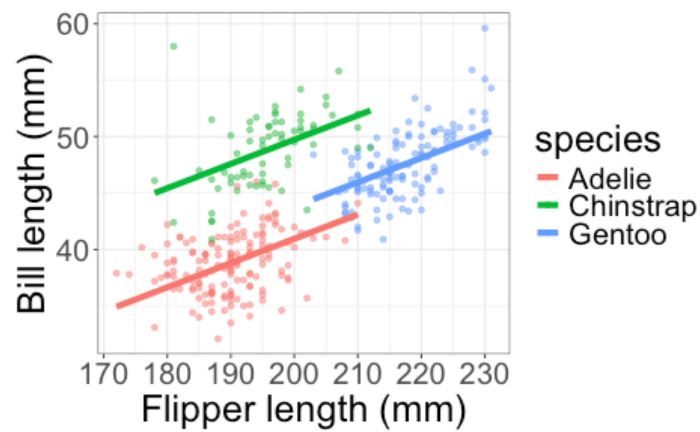
Without interaction

> Do we need to include the interaction to answer this question?

Inference

Example: Is there a difference in the slope on flipper length, for different species?





> Do we need to include the interaction to answer this question?

Yes -- we can't check if the slopes are different without fitting different slopes!

Fitting the model with interactions

```
m1 <- lm(bill_length_mm ~ species*flipper_length_mm,
         data = penguins)
```

- `*` means "include both variables *and* their interaction"

species

flipper length

species · flipper length

Class activity

https://sta112-s26.github.io/class_activities/ca_23.html

Class activity

$$\widehat{\text{Height}} = 40.689 - 7.728\ \text{IsMale} + 0.118\ \text{Age} \\ + 0.064\ \text{Age} \cdot \text{IsMale}$$

What is the estimated height for a girl aged 120 months (10 years)?

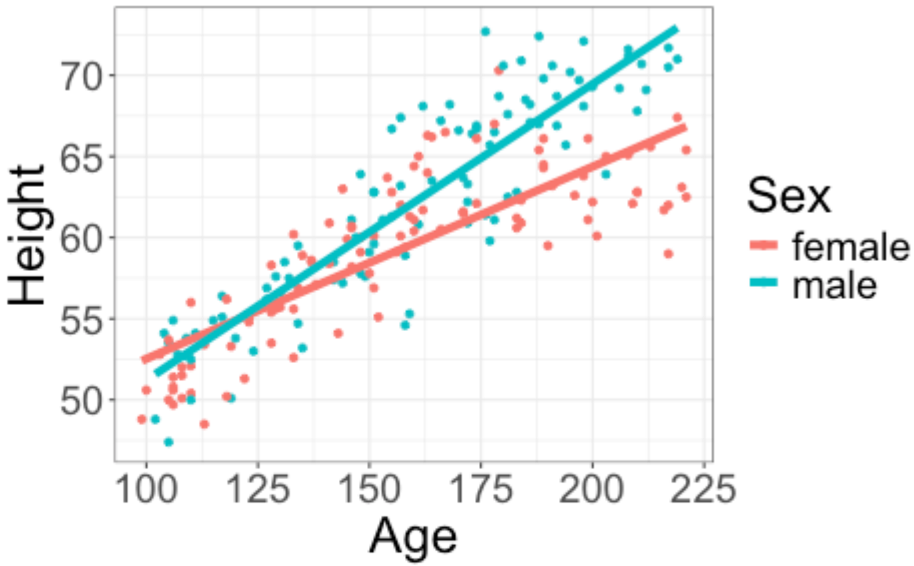
Class activity

$$\widehat{\text{Height}} = 40.689 - 7.728 \text{ IsMale} + 0.118 \text{ Age} + 0.064 \text{ Age} \cdot \text{IsMale}$$

> What is the estimated height for a girl aged 120 months (10 years)?

$40.689 + 0.118(120) = 54.85$ inches (about 4 feet 7 inches)

Class activity

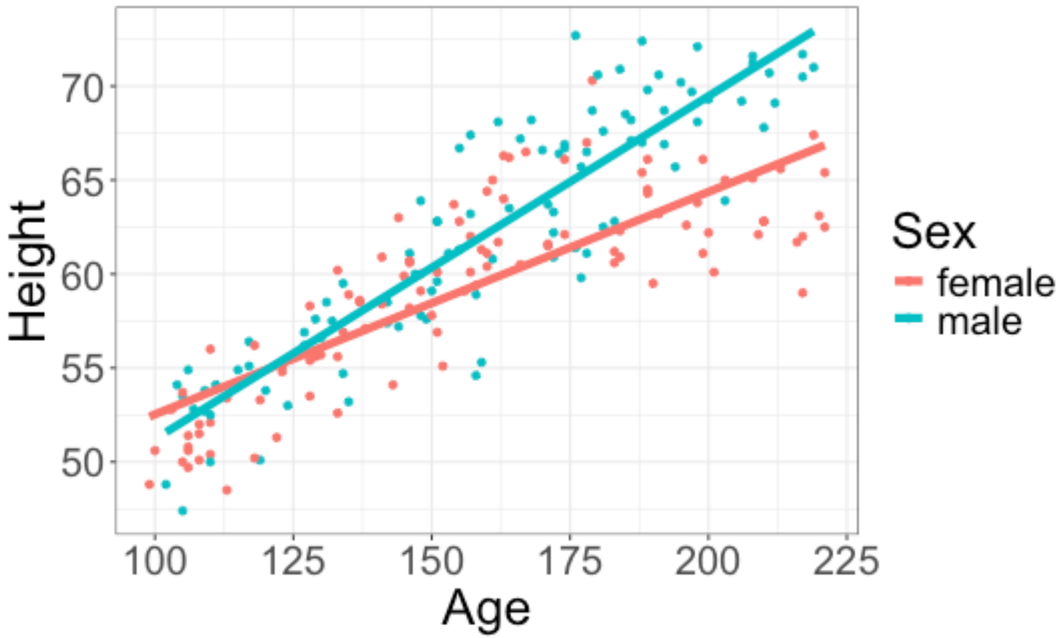


With interaction: $R^2_{adj} = 0.763$

Without interaction: $R^2_{adj} = 0.732$

> Do you think an interaction is needed?

Class activity



I think an interaction is important

- slopes look different
- captures that boys and girls start off with similar heights, but boys eventually become taller.